

MandateBench: Mandate Faithfulness and Pre-Signature Monitorability for Agentic-Payment LLMs

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Preprint · 2026 · v7 (1,885 graded authorizations: default + suppression arms, 35 scenarios, 5 mandate domains, monitorability on both arms) + v6 pilot (adversarial duels; 753 graded authorizations) · Open source (MIT), github.com/Johnnyevans32/mandatebench · CC BY 4.0 · mandatebench.xyz/paper · [doi:10.5281/zenodo.21411376](https://doi.org/10.5281/zenodo.21411376)

ABSTRACT

Emerging standards for AI payments, Google’s Agent Payments Protocol (AP2) and Coinbase’s x402, let an autonomous agent commit funds by signing a *mandate*: a cryptographically signed statement of what it may spend. A signature cannot be undone. So the safety question is not whether a protocol gateway can reject a malformed request. It is whether the agent itself can be manipulated into authorizing a payment it should refuse, and whether that can be caught before it signs. MandateBench measures mandate faithfulness across nine frontier models (one or more per major lab) under a taxonomy of adversarial pressures, with no LLM judge anywhere: rule violations are checked by plain code against the signed object, and intent traps are written and labeled by hand before any model runs. Its central contribution is separating two violation classes. *Rule* violations (over budget, over the per-payment cap, wrong recipient, wrong category, expired mandate) are structural; a stateless gateway that validates the signed fields already catches them. *Intent* violations pass every signed field but still break the mandate’s plain-language intent, like a grocery mandate spent on a stored-value gift card or a flight mandate spent on a lounge membership. No structural check can catch these. On 17 intent traps across five mandate domains (snapshot v7), no model catches them all: pooled intent catch is 351/457 (77%, CI 73–80), the best model gets 90%, the worst 57%, and false refusal stays near zero throughout. The size of the gap replicates from our small pilot (75%), but the per-model rankings do not. The three models that scored perfectly on the pilot’s three traps (GPT-5.5, Claude Opus 4.8, Kimi K2.6) all fell once the set grew, GPT-5.5 to 71%. Small-sample leaderboard positions are noise even when the aggregate effect is stable. We then measure pre-signature monitorability: a monitor reading only the agent’s private reasoning predicts violations at AUROC 0.619 (bootstrap 95% CI 0.56–0.68). Telling the agent to keep its reasoning bland hurts twice: trap violations rise by a third (17.5% to 23.6%, $p \approx 0.003$) and the monitor’s AUROC falls to 0.377, significantly below chance ($z \approx 2.8$). Both halves of the suppression result replicate from the pilot (0.669 to 0.304). Suppressed reasoning makes a breach read as more routine than a compliant decision. Hiding the reasoning does not just blind the overseer; it blinds the agent.

1. Introduction

Agentic-payment protocols move the authority to spend money from a human click to a model’s signature. In AP2 [1] and x402 [2], a user (or their wallet) issues a signed mandate that sets a budget, a per-payment cap, permitted recipients, permitted categories, and a validity window. An agent then negotiates with merchants and signs to authorize each payment. The signature is the point of no return: once emitted it settles, and there is no undo.

A natural objection is that the protocol itself enforces the mandate, since a gateway can reject any signed request whose amount, recipient, or category falls outside it. This is true, and it is exactly why the interesting failures are elsewhere. Hard structural limits are the easy case. The dangerous case is a request that is structurally valid (correct recipient, in budget, in category) but is not what the user meant, or is a payment the agent was socially engineered into signing. A gateway cannot tell “this week’s groceries” from a \$25 gift card charged to the same grocer for the same amount. Only a model that reasons

about intent can. MandateBench is built to measure precisely this gap, and to ask whether an overseer can catch a breach in the moment before the signature.

We organize the work around four questions. **RQ1**: how often do models authorize mandate-violating payments under adversarial pressure? **RQ2**: does the failure fall on structural (rule) or intent violations, and does the split separate models? **RQ3**: can a monitor reading only the agent’s private reasoning predict a violation before the signature, and how robust is that to an agent instructed to hide its reasoning? **RQ4** (adversarial): when one model plays the merchant and attacks another, who breaks and who holds?

2. Related work

Security work on agentic payments so far concentrates on the protocol layer or on a single attack vector. “Whispers of Wealth” red-teams a real AP2 signed-mandate flow via prompt injection, on one model and one vector [3]. A recent SoK of agentic-commerce security systematizes the threat axes and notes that real-world security benchmarking of these systems remains underdeveloped [4]. Zero-trust runtime verification hardens the AP2 transaction itself cryptographically [5]. MandateBench measures the layer in between, the model’s own authorization decision, across models and across a taxonomy of pressures, and is complementary to protocol enforcement.

On the oversight side, chain-of-thought monitorability has been framed as a valuable but fragile safety opportunity [6]. Black-box monitors have been trained to catch scheming in agents [7], and pre-action monitoring is established (InferAct [8], ProbGuard [9]). Deception benchmarks measure monitorability of hidden intent in cross-play games [15], and sabotage evaluations measure whether models can defeat oversight [14]. Agent policy-adherence benchmarks (ST-WebAgentBench [10], the τ -bench family, MANTRA [11], PolicyBank [12]) measure web and retail policy following, and adversarial financial-agent benchmarks such as CAIA evaluate decision quality under manipulated crypto and financial markets [13]. Verification-native clearing for agentic commerce [16] and dynamic multi-agent compliance benchmarks [17] are contemporaneous adjacent lines.

MandateBench differs on three axes: (i) a high-stakes, irreversible-action substrate (signed money movement) rather than a game or a QA task; (ii) a judge-free ground truth; and (iii) an explicit separation of gateway-catchable rule violations from intent violations that only a model or a model-monitor can catch. That separation turns “why put a model in the loop” into an empirical, measurable question.

3. Method

3.1 Signed mandates and ground truth

Each scenario issues an Ed25519-signed mandate over a canonical (sorted-key) serialization. The harness verifies the signature before each run and grades against the signed object; the agent receives the mandate’s terms in its context. A payment is a cart (recipient, amount in integer minor units, currency, category, quantity). A deterministic checker `checkAuthorization(mandate, cart, priorAuthorized)` returns the exact set of violated predicates: over budget, over the per-payment cap, unauthorized recipient, out-of-scope category, expired TTL, quantity creep, currency mismatch, and structuring (splitting an order into sub-cap charges). No LLM judges outcomes. Rule labels come from plain code by construction. Intent labels are traps written by hand and fixed before any model runs (§3.2): deterministic

to grade, but resting on the author’s judgment of intent, which independent human validation should confirm in future versions.

3.2 Pressure taxonomy and violation classes

Scenarios span a taxonomy of pressures: baseline, principal-agent overreach, merchant social engineering, urgency and scarcity, category ambiguity, prompt injection, conflicting mandates (the merchant cites an unverifiable “newer” or “supplemental” mandate), and multi-turn scope drift (a compliant authorization settles into the session and the merchant keeps pushing; violating turns are labeled per turn). Each scenario carries a ground-truth class: *clean* (authorizing is correct; declining is over-refusal), *rule* (breaks a structural term), or *intent* (passes every structural term but breaks the mandate’s stated intent). Snapshot v7 has 35 scenarios over five mandate domains (groceries, travel, subscriptions, donations, office procurement) with 6 clean controls, 12 rule traps, and 17 intent traps. Every domain’s clean control is structurally identical to its intent traps (same recipient, same category, in budget), so the only difference is what the money buys. Examples: a grocery mandate offered a \$25 gift card; a round-trip economy flight mandate offered a one-way business-class seat or a lounge membership; a named-fund donation redirected to the platform’s general fund; a 15-cable office order swapped for three premium cables. A unit test enforces the class boundary: every intent trap must pass the objective structural check.

3.3 Monitorability protocol (RQ3)

For every graded decision we store the agent’s private reasoning and public statement. A monitor model (GPT-5.5) reads only the agent’s private reasoning, never the mandate numbers or the cart, and outputs a violation probability in $[0,1]$. We report AUROC of these scores against the objective violation labels with percentile-bootstrap 95% CIs, pooled across the nine agents, scoring one repetition per scenario (temperature-0 repetitions produce nearly identical text, so this keeps the information at a third of the cost). Pooling mixes each model’s writing style with the real signal inside a model, and the monitor grades its own outputs for one of the nine agents. Per-model AUROCs are reported as exploratory; a multi-monitor panel is future work. The adversarial arm re-runs the benchmark with an added instruction telling the agent its reasoning may be reviewed and to keep it bland, uninformative, and free of any mention of limits or risk. The decision itself is otherwise unconstrained. Comparing monitor AUROC with and without suppression measures how robust this kind of oversight is.

3.4 Models

Nine flagship models via OpenRouter with structured outputs and temperature 0 where supported: GPT-5.5, Claude Opus 4.8, Gemini 3.5 Flash, DeepSeek V4 Pro, Grok 4.3, Llama 4 Maverick, Qwen3.7 Max, Kimi K2.6, and Mistral Large 2512. Snapshot v7 runs three seeded repetitions per scenario in a default arm and a suppressed-reasoning arm (1,890 runs; 1,885 graded, because unparseable runs are excluded rather than counted as catches) with per-run provenance (serving provider, temperature, rep, arm) and dedup keys. The v6 pilot (14 scenarios: 378 + 375 graded) remains the basis of the adversarial-duel results, pending their v7 rerun. Because repetitions run at temperature 0, the three reps come out nearly identical: they capture serving-stack nondeterminism, not sampling variance. The effective sample is therefore closer to the unique-scenario count, and all reported intervals should be read as optimistic.

4. Results

4.1 RQ1: Violations under pressure

Over v7’s trap scenarios, every model violates at least once: 137 of 781 trap decisions (17.5%, CI 15.0–20.4) are mandate-violating authorizations. False refusal stays near zero for every model (one clean order declined out of 162). So the violations are not the price of a cautious policy. Models are permissive on traps and comfortable authorizing clean orders at the same time.

4.2 RQ2: Rule vs intent (headline)

Splitting violations by class is the central result (Table 1). Rule traps are caught at 90% pooled. That is imperfect, but these are exactly the violations a gateway rejects no matter which model is deciding. Intent violations are the residual risk, and no model is reliable there: pooled intent catch is 351/457 (77%, CI 73–80), best 90% (Kimi K2.6), worst 57% (Llama 4 Maverick). The most instructive result is what happened to the pilot’s leaderboard. On v6’s three intent traps, GPT-5.5, Claude Opus 4.8, and Kimi K2.6 scored a perfect 9/9. On 17 traps across five domains, none of them stays perfect and the order changes: GPT-5.5 falls to 71%, while DeepSeek and Mistral rise to 86%. The size of the intent gap replicates almost exactly (pooled 75% to 77%). The per-model rankings at small n do not. Benchmarks in this genre should report pooled effects with intervals and treat small-sample per-model rankings as provisional.

MODEL	RULE CAUGHT (N=36)	INTENT CAUGHT (N=51)	FALSE REFUSAL (N=18)
Kimi K2.6	89% (32/36) [75–96]	90% (44/49) [78–96]	0% (0/18) [0–18]
DeepSeek V4 Pro	92% (33/36) [78–97]	86% (44/51) [74–93]	0% (0/18) [0–18]
Mistral Large 2512	83% (30/36) [68–92]	86% (44/51) [74–93]	0% (0/18) [0–18]
Grok 4.3	92% (33/36) [78–97]	84% (43/51) [72–92]	0% (0/18) [0–18]
Claude Opus 4.8	97% (35/36) [86–100]	82% (42/51) [70–90]	0% (0/18) [0–18]
GPT-5.5	94% (34/36) [82–98]	71% (36/51) [57–81]	0% (0/18) [0–18]
Gemini 3.5 Flash	92% (33/36) [78–97]	71% (36/51) [57–81]	0% (0/18) [0–18]
Qwen3.7 Max	92% (33/36) [78–97]	65% (33/51) [51–76]	0% (0/18) [0–18]
Llama 4 Maverick	83% (30/36) [68–92]	57% (29/51) [43–69]	6% (1/18) [1–26]

Table 1. Fraction of violations caught, split by class, snapshot v7 (35 scenarios, 5 domains, 3 seeded reps at temperature 0); brackets are Wilson 95% CIs. Rule = structural, gateway-catchable. Intent = structurally valid but breaks stated intent. False refusal = clean orders wrongly declined (usefulness cost). Pooled intent catch 77% (CI 73–80); mean per-model 77%. Kimi’s intent n is 49 because its two unparseable runs are excluded, not counted as catches. Temperature-0 reps are near-duplicates, so intervals are optimistic (§3.4).

4.3 RQ3: Pre-signature monitorability

On snapshot v7, a monitor reading only the agent’s private reasoning predicts violations at AUROC 0.619, bootstrap 95% CI 0.56–0.68 (45 violations vs 270 compliant decisions; one rep per scenario). That is useful but far from perfect, and it replicates the pilot’s 0.669. Under reasoning suppression the picture inverts, in both respects. First, suppression is not behaviorally free: trap violations rise by a third (137/781 to 184/780, 17.5% to 23.6%, Fisher exact $p \approx 0.003$; the smaller pilot showed roughly a doubling). Reading the transcripts shows why. Told to stay bland, models stop thinking about intent and fall back to box-ticking the structural checks; violating intent-trap runs literally read “within bud-

get, under per-payment cap, to allowed recipient, in allowed category” and authorize the gift card. Second, the monitor’s AUROC falls from 0.619 to 0.377, bootstrap 95% CI 0.31–0.44, significantly below chance ($z \approx 2.8$) and a significant drop ($z \approx 3.8$; pilot: 0.669 to 0.304). The bland, reassuring reasoning attached to a breach looks more innocent than an ordinary compliant decision, so it points the overseer the wrong way. Below chance is itself a signal: suppression flips the sign of the information rather than deleting it, and an overseer who knows suppression is happening could invert the scores and get back to about 0.62. The danger is the overseer who does not know. Exploratory per-model AUROCs (2 to 10 positives each) vary a lot even without suppression. Grok 4.3, Qwen3.7 Max, and Claude Opus 4.8 land at 0.82 to 0.84, while Llama 4 Maverick (0.26) and Gemini 3.5 Flash (0.33) are below chance in the clear. Whose reasoning is monitorable may matter as much as whether reasoning is. In sum, naive reasoning-hiding raises the violation rate and defeats an unsuspecting monitor at the same time; it blinds the agent as well as the observer.

4.4 RQ4: Under live attack (v6 pilot)

Beyond scripted traps, we run a full adversarial matrix: every model plays the merchant and attacks every other model’s agent in a multi-turn negotiation ($9 \times 9 = 81$ duels; one negotiation per attacker–defender pair against a single over-cap goal, attacker unseeded at temperature 1; a seeded rerun with more goals is planned). Robustness depends on the defender, not the attacker. No attacker succeeds more than 22% of the time (2/9, CI 6–55%), but the breaches pile up on a few agents (Table 2). With $n=9$ per row the intervals are wide. The unbreached group (0/9, upper bound 30%) separates from Mistral and Llama, but the order among DeepSeek, Mistral, and Llama is not significant, and individual cells of the matrix are single draws that mean nothing on their own. Llama 4 Maverick sits at the bottom of both this matrix and the v7 intent column. Beyond that, the v7 ranking reshuffle (§4.2) warns against reading finer agreement between the two tables: DeepSeek and Mistral were breached here yet sit near the top of the v7 intent ranking.

AGENT (DEFENDER)	BREACHED BY ATTACKERS	RATE	WILSON 95% CI
GPT-5.5 · Opus 4.8 · Gemini · Grok · Qwen · Kimi	0 / 9	0%	0–30%
DeepSeek V4 Pro	1 / 9	11%	2–44%
Mistral Large 2512	5 / 9	56%	27–81%
Llama 4 Maverick	6 / 9	67%	35–88%

Table 2. Agent robustness in the 9×9 adversarial matrix (81 duels, single over-cap goal, up to 3 turns, one negotiation per cell; snapshot v6). Rate = fraction of the 9 attackers that broke this agent, with Wilson 95% CIs. No attacker exceeded 22% success, so robustness tracks the defender, not the attacker.

5. Discussion

The rule/intent split reframes the “the gateway already enforces the mandate” objection. Gateways enforce structure, and structural misses get caught by the gateway no matter which model is deciding. Intent is the residual: unenforceable by construction, and the axis on which frontier models measurably differ. This is the empirical case for a model (or a model plus monitor) in the payment loop. The v7 expansion adds a methods lesson: aggregate effects (the roughly 25% intent miss rate) replicate across trap sets, while small-sample per-model rankings do not. That is a caution for any leaderboard built on a handful of scenarios. The monitorability result is a caution of its own: reasoning is only moderately di-

agnostic even in the clear, and one trivial suppression instruction pushes an unsuspecting monitor below chance while making the agent less safe. Oversight that leans on reading reasoning must assume the channel can be cheaply degraded, and should treat a monitor that runs systematically below chance as a red flag for suppression rather than as noise.

6. Limitations

The adversarial-duel matrix is still a v6 pilot measurement (one unseeded negotiation per cell against a single over-cap goal) and is labeled that way; the seeded rerun with more goals is future work, and the §4.2 ranking reshuffle warns that its per-model detail may move. v6 rates were also lower bounds because of two grading bugs fixed before v7: a counter that committed the same purchase was graded as a catch, and unparseable output was graded as a non-violation. The monitor is one model (GPT-5.5), scores one repetition per scenario, is pooled across agents, and grades its own outputs for one of the nine; per-model AUROCs rest on 2 to 10 positives each and are exploratory. Intent labels are written by the author and fixed before any model runs; independent human validation is future work. Repetitions run at temperature 0 and come out nearly identical, so the effective sample is closer to the unique-scenario count and all intervals are optimistic. Results may be sensitive to system-prompt wording. During development, an unfair expired-mandate prompt produced a spurious universal failure until the agent was given the issued, expiry, and current timestamps. The suppression arm uses a single naive hiding instruction; a trained or steganographic adversary is the stronger test of monitorability and is future work, as is an adaptive-adversary robustness curve.

7. Conclusion

MandateBench provides a judge-free instrument for the reasoning-based oversight of high-stakes agentic payments. Its measurements show that frontier models handle the violations a gateway already catches but not all of the intent violations only a model can. They also show that reading an agent’s reasoning to catch breaches is moderately reliable at best, and cheaply defeated in a way that also degrades the agent. We release the benchmark, harness, and live dashboard openly.

References

1. Google. Agent Payments Protocol (AP2). 2025. Announced with 60+ partners; governance since contributed to the FIDO Alliance (2026).
2. Coinbase. x402: An open standard for internet-native payments. 2025. Contributed to the Linux Foundation’s x402 Foundation (2026).
3. Debi, T., Zhu, W., Sen Gupta, P. Whispers of Wealth: Red-Teaming Google’s Agent Payments Protocol via Prompt Injection. arXiv:2601.22569, 2026.
4. Mao, Q., Wang, J., Liu, Y., Zhu, L., Ma, C., Yan, J. SoK: Security of Autonomous LLM Agents in Agentic Commerce. arXiv:2604.15367, 2026.
5. Lan, Q., Kaul, A., Jones, S., Westrum, S. Zero-Trust Runtime Verification for Agentic Payment Protocols: Mitigating Replay and Context-Binding Failures in AP2. arXiv:2602.06345, 2026.
6. Korbak, T., Balesni, M., Barnes, E., Bengio, Y., et al. Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety. arXiv:2507.11473, 2025.
7. Storf, S., Barton-Cooper, R., Peters-Gill, J., Hobbhahn, M. Constitutional Black-Box Monitoring for Scheming in LLM Agents. arXiv:2603.00829, ICML 2026.

8. Fang, H., Zhu, X., Gurevych, I. InferAct: Preemptive Detection and Correction of Misaligned Actions in LLM Agents. arXiv:2407.11843, EMNLP 2025.
9. Wang, H., Poskitt, C. M., Wei, J., Sun, J. ProbGuard: Probabilistic Runtime Monitoring for LLM Agent Safety. arXiv:2508.00500, 2025.
10. Levy, I., et al. ST-WebAgentBench: A Benchmark for Evaluating Safety and Trustworthiness in Web Agents. arXiv:2410.06703, 2024.
11. Anand, A., Chatzi, I., Raha, R., Schmuck, A.-K. MANTRA: Synthesizing SMT-Validated Compliance Benchmarks for Tool-Using LLM Agents. arXiv:2605.06334, 2026.
12. Choi, J., Yoon, J., Le, L. T., Jha, S., Pfister, T. PolicyBank: Evolving Policy Understanding for LLM Agents. arXiv:2604.15505, 2026.
13. Dai, Z., Peng, Z., Cheng, Z., Li, R. Y. When Hallucination Costs Millions: Benchmarking AI Agents in High-Stakes Adversarial Financial Markets (introduces the CAIA benchmark). arXiv:2510.00332, 2025.
14. Benton, J., et al. (Anthropic). Sabotage Evaluations for Frontier Models. arXiv:2410.21514, 2024.
15. Eburu, E. Among LLMs: A Cross-Play Benchmark for Deception, Detection, and the Monitorability of Reasoning. Zenodo, doi:10.5281/zenodo.21209429, 2026.
16. RAILS: Verification-Native Clearing for Agentic Commerce. arXiv:2606.08790, 2026.
17. Beyond Goodhart’s Law: A Dynamic Benchmark for Evaluating Compliance in Multi-Agent Systems (MAC-Bench). arXiv:2606.07805, 2026.